CloudFront Geographic Restrictions using EC2 Origin

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon EC2** instance was configured as a web server and used as an origin for an **Amazon CloudFront** distribution. Unlike previous labs that used **Amazon S3** as the origin, this lab demonstrated how to use an EC2-hosted website by configuring Apache HTTP Server and deploying a custom web page. After creating the CloudFront distribution, **CloudFront Geographic Restrictions** were configured to block website access from selected countries. The restrictions were tested using a geo-location testing tool and by accessing the website from India. Finally, the restrictions were removed and normal access was restored.

Understanding Geographic Restrictions in Amazon CloudFront

CloudFront Geographic Restrictions allow content owners to control which countries can access their CloudFront distributions. CloudFront determines the viewer's country using the source IP address and then either allows or blocks access based on configured rules. This feature is commonly used for licensing requirements, regulatory compliance, regional content distribution, and restricting unwanted traffic. Geographic restrictions can be configured using either an allow list or a block list.

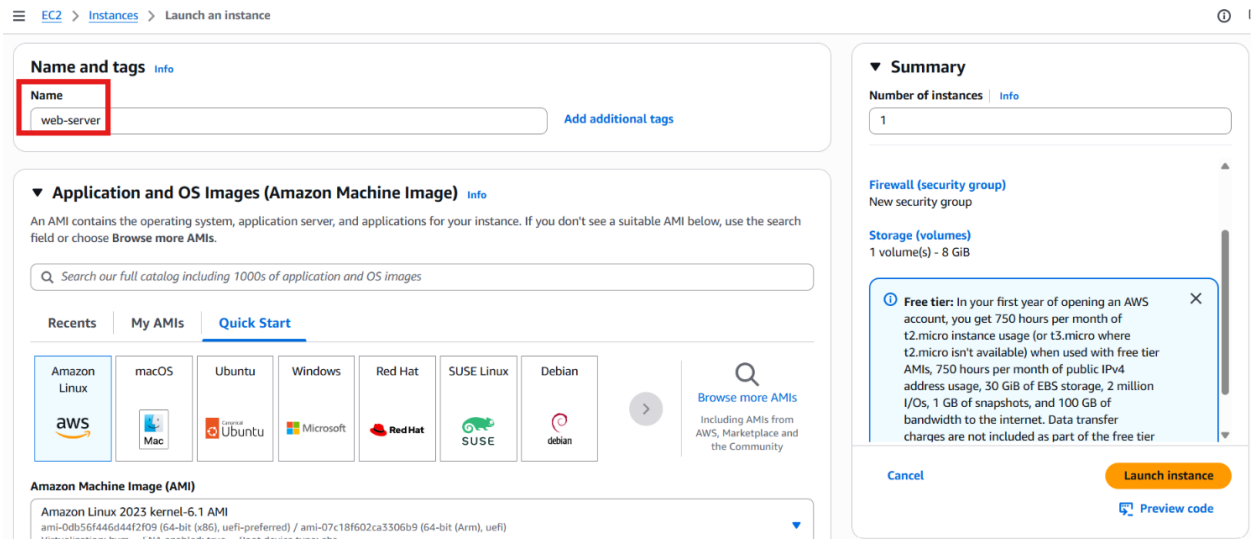
Example:

A video streaming company owns distribution rights only in the United States and Canada. Using CloudFront Geographic Restrictions, the company allows access only from those countries and automatically blocks viewers from all other regions.

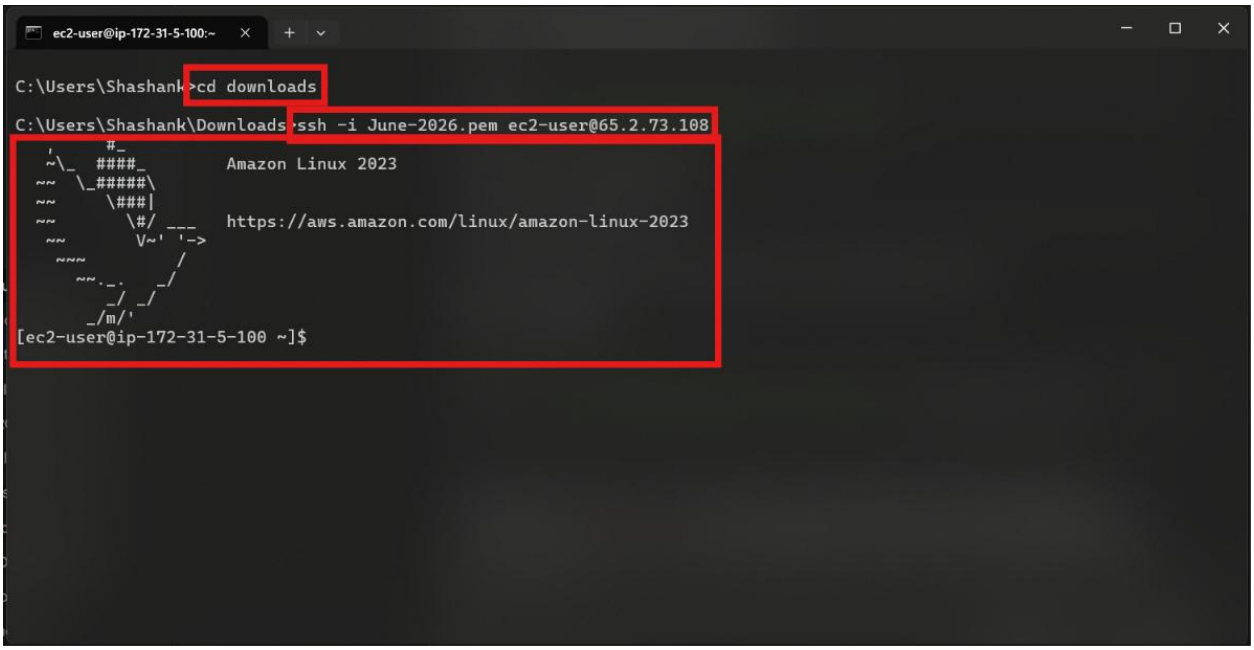
Lab Steps

Step 1 — Create and Configure the EC2 Web Server

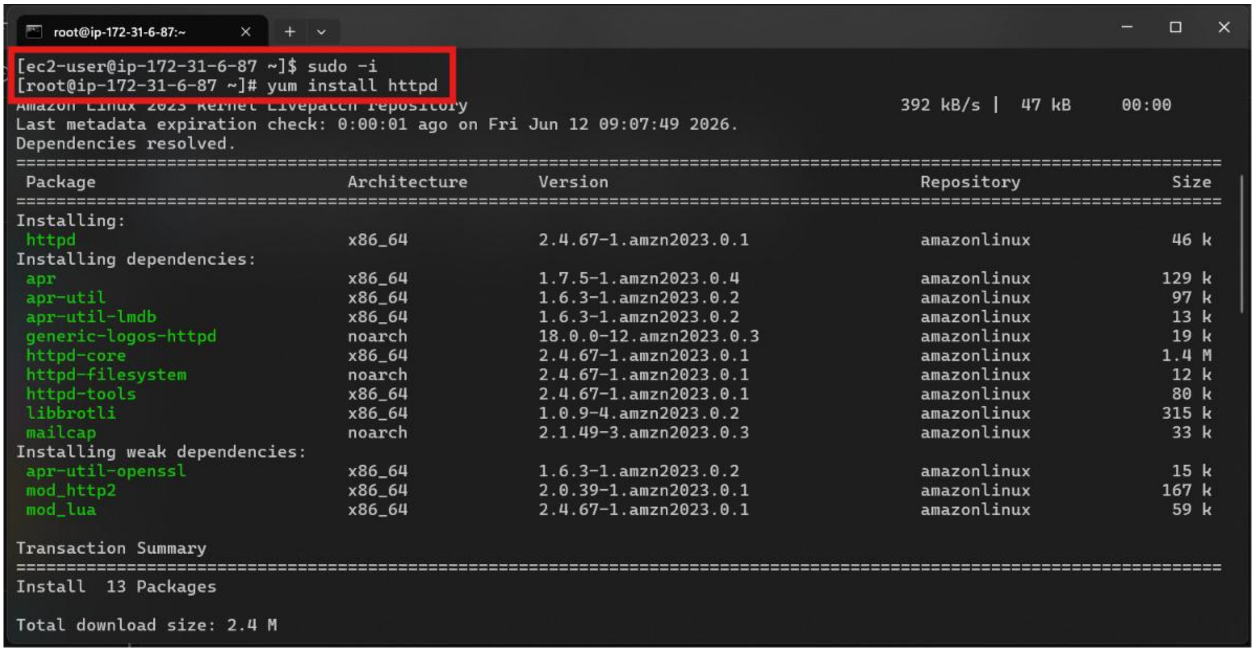
- Create an **Amazon EC2** instance and configure name as **web-server** and enable **Allow HTTP traffic from the internet.**



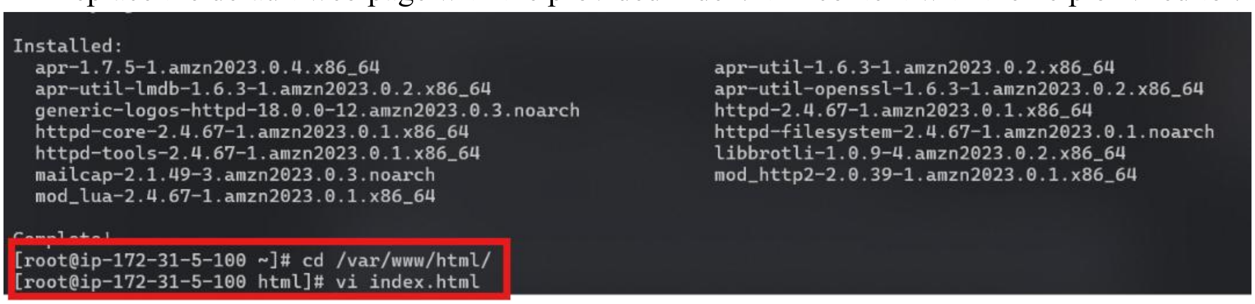
- Now we will configure it as a web server so we first must access the instance.
- For this, we will use **Command Prompt** and run the following commands.
- Now type: **cd {location_of_your_security_key}.**
- Now type: **ssh -i{security_key_name} ec2-user@{public_ipv4_address_of_ec2_instance}.**



- Now login as root user by typing this command: **sudo -i.**
- Use **yum install httpd** command to configure and place webpage in Apache on the EC2 instance.



- Replace the default web page with the provided index.html content with the help of vi editor.



- Use the above commands to change content of the default index.html and paste your code like this or use the code below.

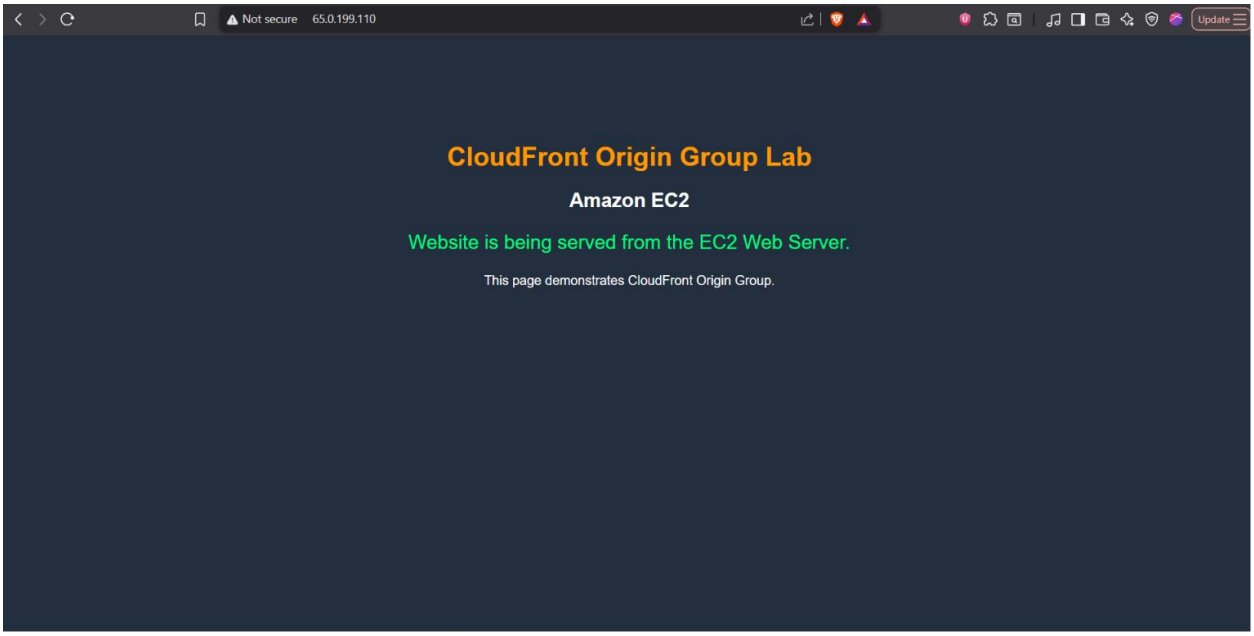
```
<!DOCTYPE html>
<html>
<head>
   <title>CloudFront Origin Group Demo</title>
   <style>
      body {
         font-family: Arial, sans-serif;
         background-color: #232F3E;
         color: white;
         text-align: center;
         padding-top: 100px;
      }
      .container {
         width: 70%;
```

```
        margin: auto;
    }
    h1 {
        color: #FF9900;
    }
    .status {
        font-size: 24px;
        color: #00FF7F;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>CloudFront Origin Group Lab</h1>
    <h2>Amazon EC2</h2>
    <p class="status">Website is being served from the EC2 Web Server.</p>
    <p>This page demonstrates CloudFront Origin Group.</p>
  </div>
</body>
</html>
```

- Now enable and run the web server services using these commands

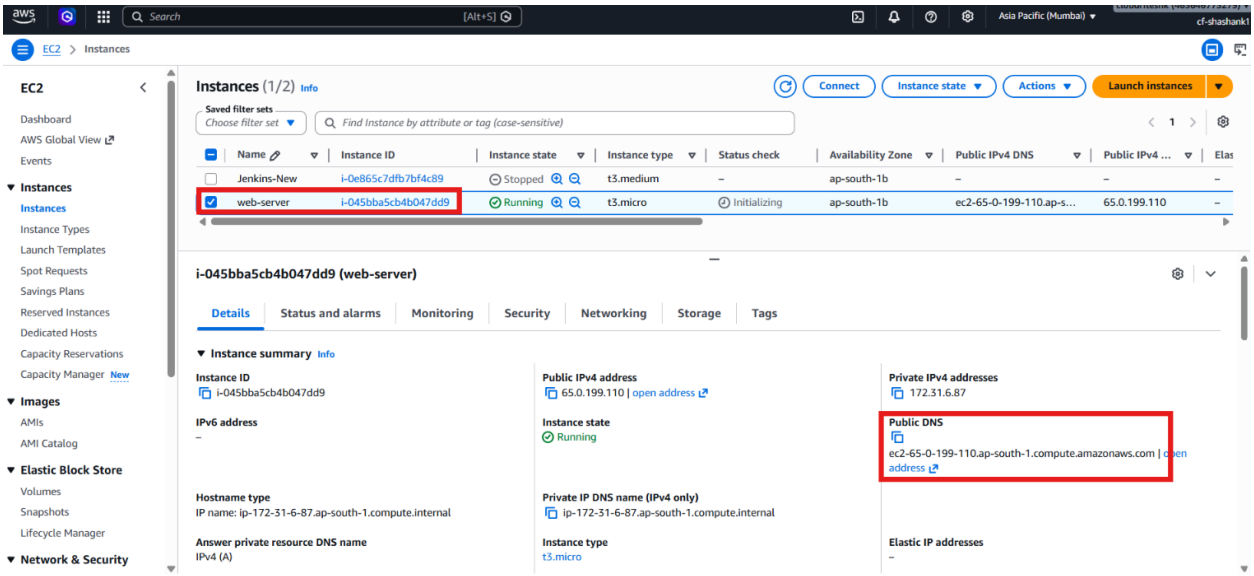
```
systemctl start httpd
systemctl enable httpd
```

- Verify that the EC2 website opens correctly using the public IP.

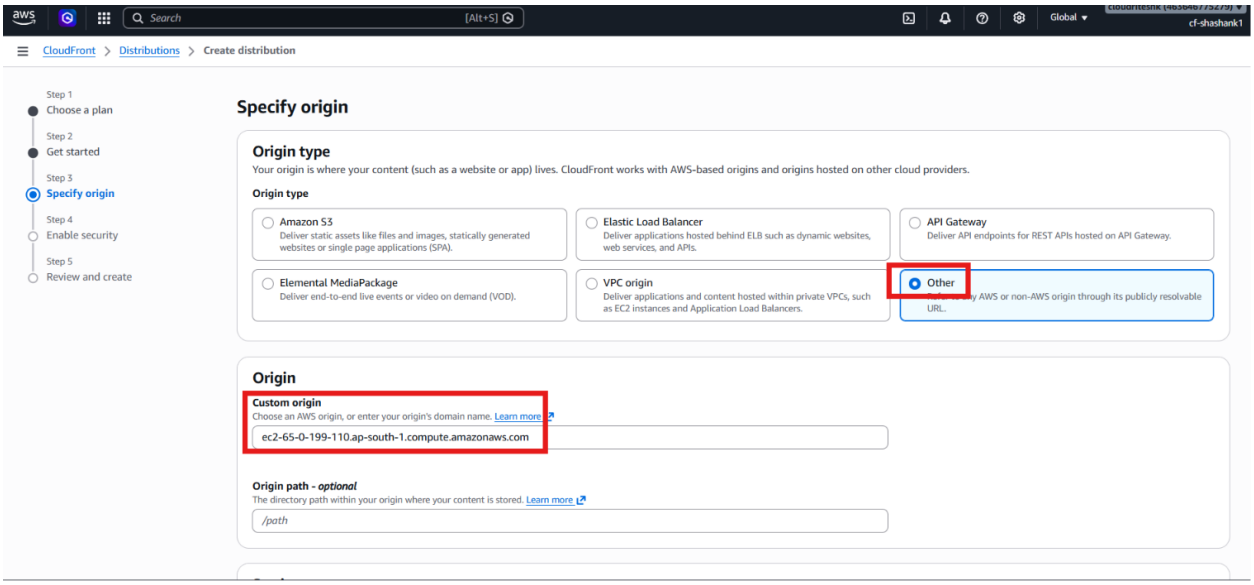


Step 2 — Create the CloudFront Distribution

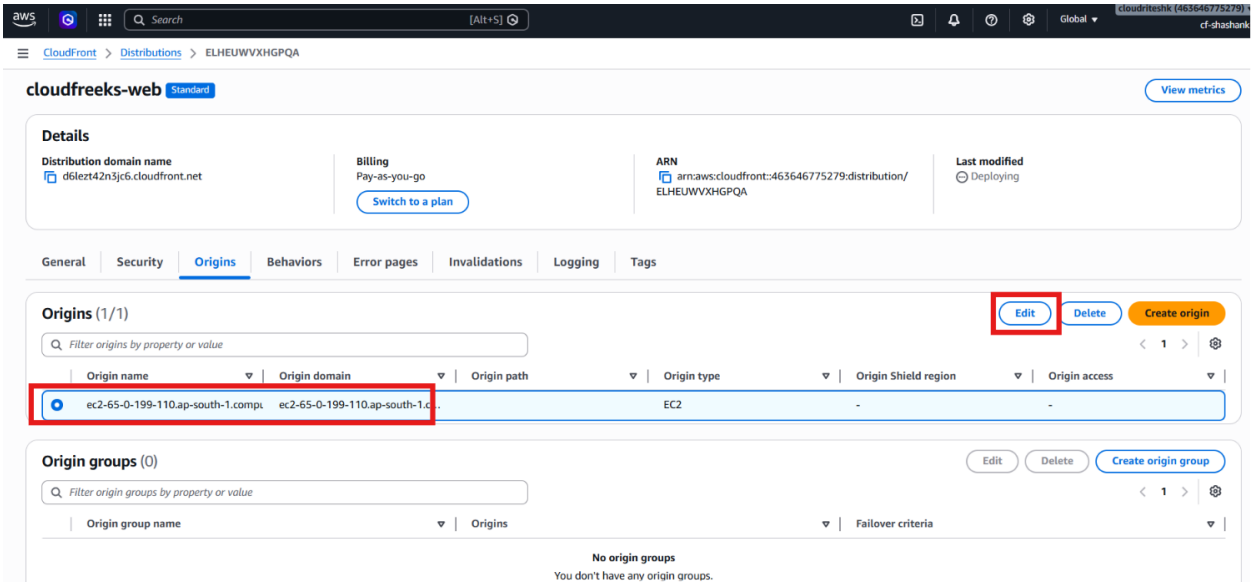
- Navigate to **Amazon CloudFront**.
- Click **Create Distribution**.
- Copy the EC2 instance **Public IPv4 DNS** value.



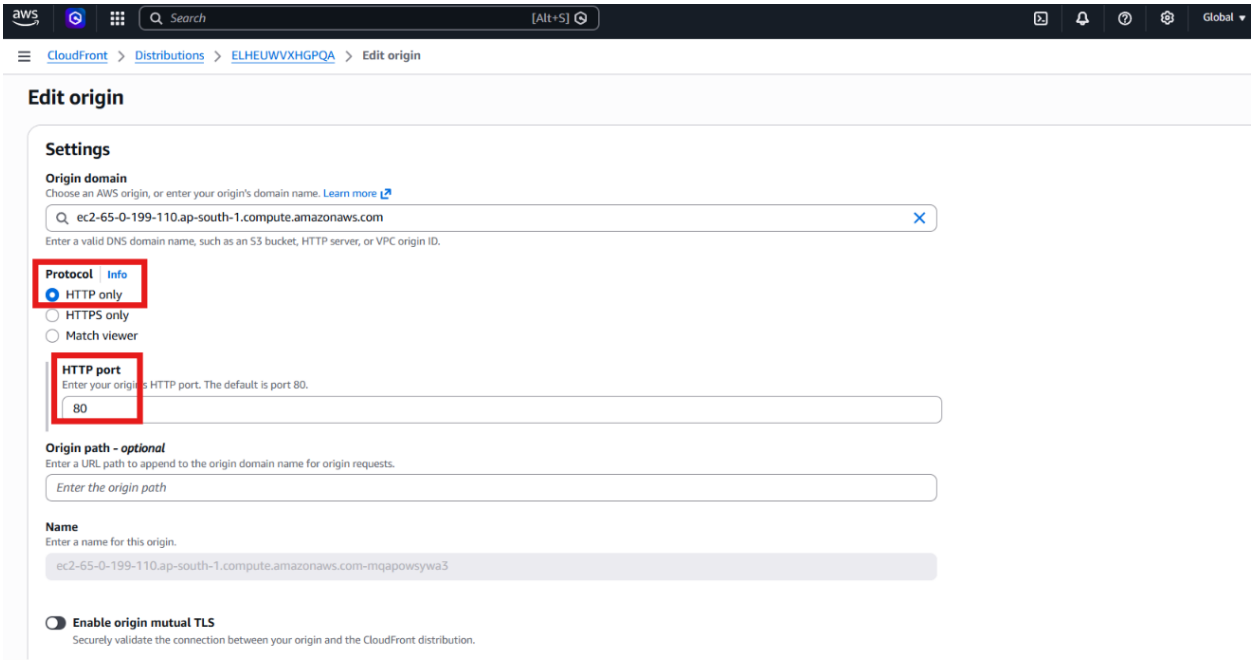
- Paste the DNS name into the **Origin Domain** field after selecting type as **Other** and create the distribution.



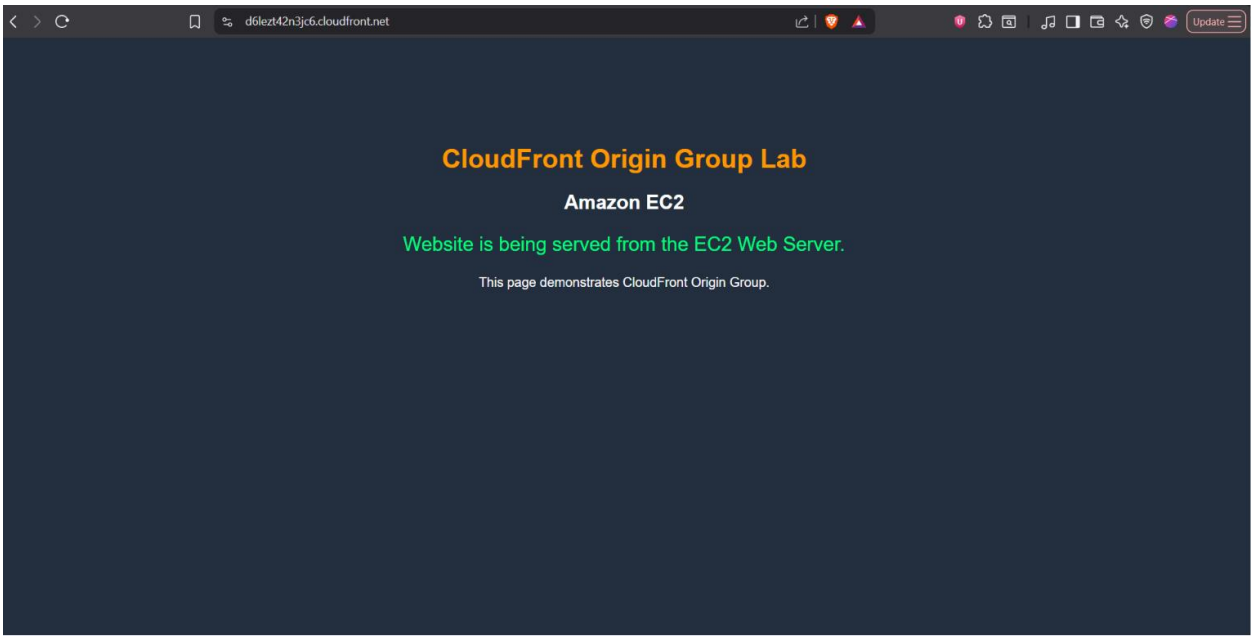
- Edit the Origin.



- Configure the origin protocol as **HTTP Only** and **Port** as **80.**

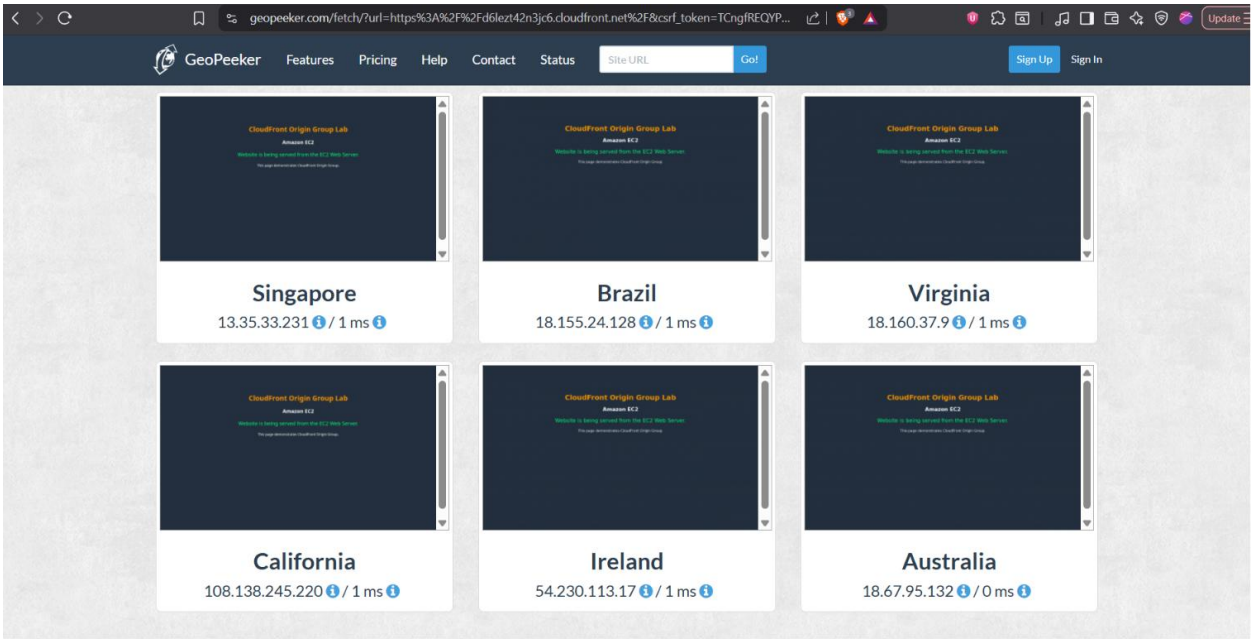


- Wait for the distribution deployment to complete.
- Verify the website opens successfully through the CloudFront URL.



Step 3 — Test Global Website Accessibility

- Copy the CloudFront distribution URL.
- Open the website through the CloudFront URL.
- Use a geographic testing tool such as GeoPeeker to test access from multiple countries.
- Verify that the website is accessible from different regions including Australia, Ireland, Singapore, and Brazil.

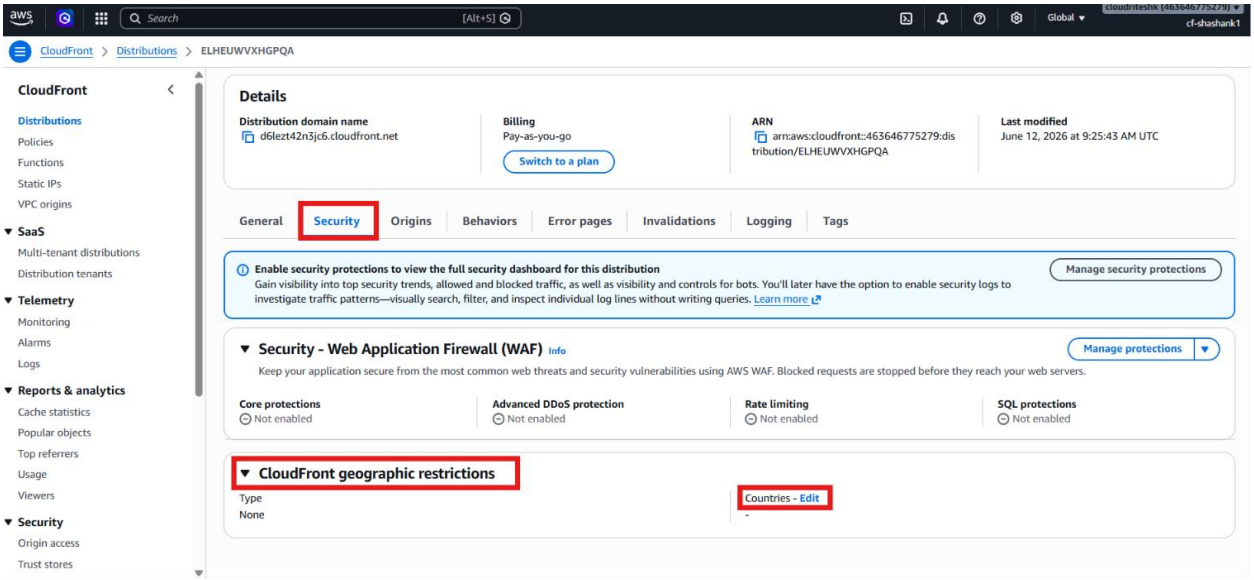


- Confirm that no geographic restrictions are currently configured.

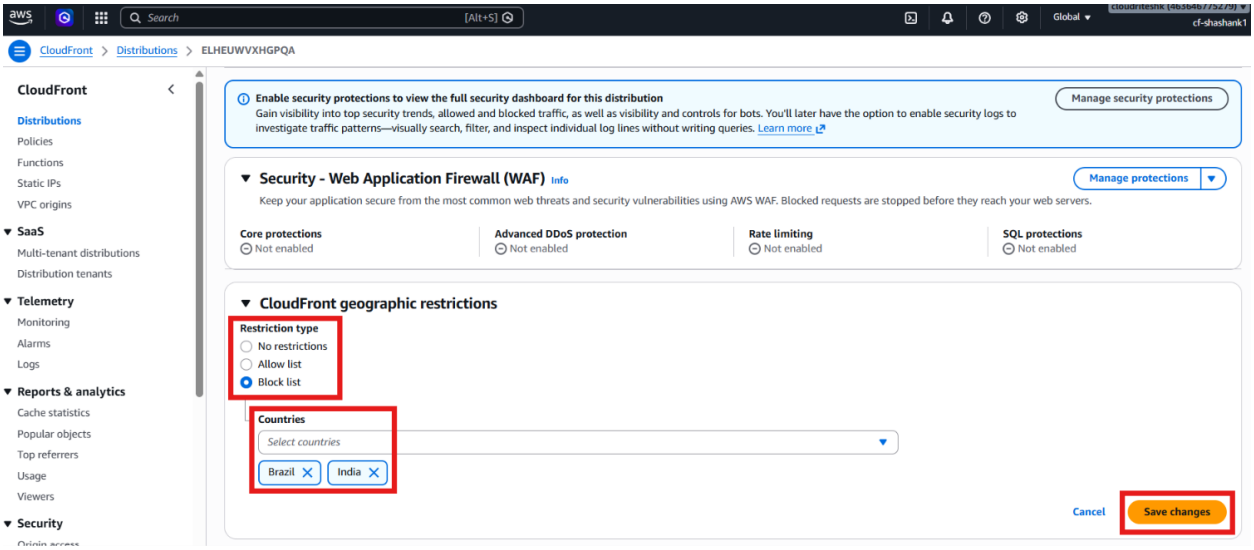
Step 4 — Configure Geographic Restrictions

Configuration Steps

- Open the CloudFront distribution.
- Navigate to **Security**.
- Locate **Geographic Restrictions**.
- Select **Countries - Edit**.



- Add **India**.
- Add **Brazil**.
- Save the configuration.

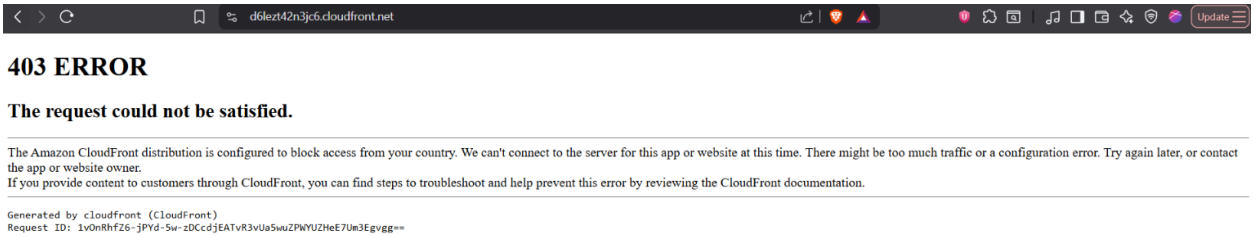


- Wait for CloudFront to redeploy the distribution.

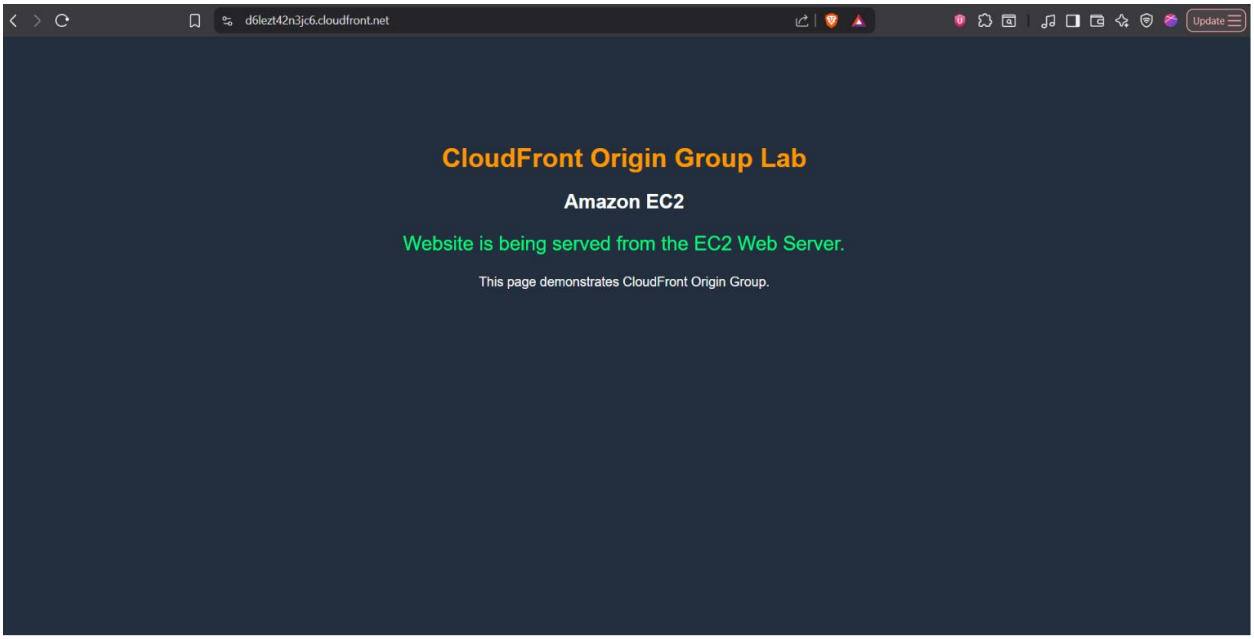
Step 5 — Verify and Remove Geographic Restrictions

- Open the CloudFront URL from India.

- Verify that CloudFront returns a **403 Forbidden** error.



- Confirm that the access denied message indicates the distribution is configured to block access from the country.
- Use GeoPeeker to verify that Brazil is also blocked.
- Confirm that countries not included in the block list can still access the website.
- Return to **CloudFront → Security → Edit**.
- Change Geographic Restrictions to **None**.
- Save the changes.
- Wait for CloudFront deployment to complete.



- Verify that the website becomes accessible again from India and Brazil.

Verification

- EC2 instance successfully hosts the website using Apache HTTP Server.
- CloudFront distribution successfully uses the EC2 instance as its origin.
- Website is accessible globally before restrictions are applied.
- Geographic restrictions successfully block access from India and Brazil.
- CloudFront returns a 403 error for blocked countries.
- Removing the restriction restores normal website access.
- Geographic filtering is successfully implemented using **Amazon CloudFront Geographic Restrictions**.